Pressure correction in free-surface flow

2D Truchas working note

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Purpose and scope

This note compares the pressure-correction paths currently used by the two-dimensional flow code and by mainline Truchas. The comparison is focused on the cell-centered velocity correction in a fluid/VOID calculation. The two implementations use closely related pressure systems and face-centered corrections; the important difference is how the pressure correction is converted into a cell-centered velocity.

The issue was exposed by a translating fluid plug on a perturbed mesh at a small Courant number. A mixed fluid/VOID cell retained a very small amount of real fluid. The pressure solve converged, but the resulting cell-centered velocity correction became enormous.

Discrete projection setting

Let c denote a cell and f a face. The projection computes a pressure correction δp_c from a discrete incompressibility equation

$$A\,\delta p = b,$$

where A is assembled from face coefficients. In both implementations the coefficient used on a face is the inverse of a face density,

$$\alpha_f = \frac{1}{\rho_f}.$$

Schematically, the matrix is the finite-volume composition

$$A \simeq -D\,M_f(\alpha_f)\,G,$$

with G a pressure-to-face-gradient operator, M_f the geometric face scaling, and D a face-flux divergence operator. The exact geometric weights are implementation-specific, but the important point is that the pressure system is face-based and uses α_f.

The pressure correction is then used in two places:

1. to correct the face-normal velocity used by the divergence constraint; and
2. to update the cell-centered velocity used by the rest of the flow algorithm and by output.

The first operation is the one that directly enforces the projection. The second should represent the same physical correction in a cell-centered form.

Common face correction

Both codes apply the face correction in the form

$$u_f^{n+1} = u_f^* - \Delta t\, \alpha_f\, (G_f\, \delta p)_f,$$

where u_f^* is the predicted face-normal velocity and G_f includes the boundary-condition treatment for pressure correction. In the 2D code this is the pressure_derivative/inv_density_f path. In mainline it is the velocity_fc_correct path using gradient_cf and rho_fc.

Consequently, the pressure system and the face velocity correction already use the same basic design in the two codes. The observed instability is not evidence by itself that the pressure matrix is nonsymmetric or that the projection solve failed: in the diagnostic run the projection residual was about 10^{-12}, and the solve reported success.

The current 2D cell-centered correction

The current 2D projection update stores a cell-centered pressure-gradient-like quantity at the old and new states. Suppressing boundary and gravity details, the update has the form

$$u_c^{n+1} = u_c^* - \Delta t\, \alpha_c \left[(\nabla_c p)^{n+1} - (\nabla_c p)^n \right],$$

where

$$\alpha_c = \frac{1}{\rho_c}$$

and ∇_c is the 2D cell-centered gradient_cc operator. In the source, the operation is represented by gradient_pressure followed by a direct multiplication by inv_density_c.

This is a reasonable approximation for a single-density fluid. If density is smooth and bounded away from zero, the distinction between

$$\frac{1}{\rho_c} \nabla_c p \quad \text{and} \quad I_{fc}\left(\frac{1}{\rho_f} G_f p \right)$$

is an ordinary discretization difference. For a fluid/VOID interface, however, ρ_c represents the mass density averaged over the active cell volume. A cell containing a small real-fluid fraction θ can therefore have, schematically,

$$\rho_c \simeq \theta\, \rho_{fluid}, \qquad \alpha_c \simeq \frac{1}{\theta\, \rho_{fluid}}.$$

The cell-centered correction then amplifies an otherwise modest pressure gradient by $1/\theta$.

The current free-surface diagnostic gives a concrete example:

$$\theta \simeq 3.78 \times 10^{-8}, \qquad \alpha_c \simeq 2.65 \times 10^{7}.$$

At the first anomalous cell, the projection converged with a relative residual of approximately 9.9×10^{-13}, but the cell-centered correction was approximately

$$(20.17,\ 1.069).$$

The large velocity is thus a correction-amplification problem after a successful pressure solve, not a failure of the linear solver to reduce its residual.

Mainline's face-density-scaled cell correction

Mainline forms a pressure-gradient quantity already divided by face density. For each active face it computes, schematically,

$$q_f = \frac{1}{\rho_f} \left(G_f p + h_f \right),$$

where h_f represents the gravity-head contribution when gravity is active. It then interpolates that face quantity to the cell center,

$$q_c = I_{fc} q_f,$$

and updates the cell-centered velocity as

$$u_c^{n+1} = u_c^* - \Delta t \left(q_c^{n+1} - q_c^n \right).$$

There is no additional multiplication by $1/\rho_c$ in this final update. In the source this is the sequence grad_p_rho, interpolate_fc, and velocity_cc_correct.

Mainline also constructs ρ_f from neighboring cell properties using fluid-volume weighting and enforces a positive face-density floor. In schematic form,

$$\rho_f = \max\left(\rho_{f,\mathrm{weighted}}, \rho_{f,\mathrm{min}} \right).$$

The floor is based on the minimum real-fluid density and a configurable face fraction. Therefore a tiny fluid fragment does not produce an arbitrarily large inverse density in the pressure correction used by the face and cell-centered velocity paths.

Comparison

The essential difference can be summarized as follows.

Aspect	Current 2D path	Mainline path
Projection matrix	Face-based; uses $1/\rho_f$	Face-based; uses $1/\rho_f$
Face velocity correction	Face gradient divided by face density	Face gradient divided by face density
Cell velocity correction	Cell gradient multiplied by $1/\rho_c$	Face gradient divided by ρ_f, then interpolated to cells
Small fluid/VOID fraction	Can make $1/\rho_c$ very large	Controlled by face-density weighting and floor
Primary consistency	Cell and face paths use different density/gradient pairings	Cell path is derived from the same face-scaled quantity as the face path

For a full fluid cell with smooth density, the two cell-centered formulas are close and may be indistinguishable at the expected discretization error. At a free surface, they are not equivalent: the current 2D formula applies a cell-centered inverse density after the gradient has been formed, while the mainline formula applies the density scaling on faces before interpolation.

Cell classification is not the root cause

The problematic cell is a mixed fluid/VOID cell adjacent to pure VOID. Both implementations classify this configuration as an active regular cell rather than simply deleting it. The reason is that a mixed cell can still exchange fluid with its neighboring active cells, and the interface reconstruction needs to retain it.

Thus changing the classification alone would be a poor fix. Promoting the cell to VOID would discard a geometrically represented fluid fragment; keeping it active is consistent with the mainline classification contract. The important question is how pressure and velocity corrections are scaled in that active cell.

Implications for the 2D implementation

The most direct alignment with mainline is to replace the current cell correction by a 2D analogue of the face-density-scaled path:

1. compute the pressure-gradient contribution on faces;
2. divide by the bounded face density, including the free-surface floor;
3. interpolate the resulting vector to cell centers; and
4. subtract the difference between the new and old interpolated quantities.

The face correction and the projection matrix can remain as they are initially. This change targets the demonstrated amplification mechanism without changing the material tracker or silently deleting small fluid fragments.

Simply clamping inv_density_c would suppress the symptom, but it would leave the cell-centered correction mathematically disconnected from the face correction and would introduce a second, less transparent free-surface regularization. It may still be useful as a diagnostic comparison, but it is not the preferred final formulation.

Limitations and follow-up questions

The face-density-scaled correction resolves the immediate mismatch between the 2D and mainline formulations, but it does not by itself settle every free-surface question. In particular, future work should verify:

- the appropriate face-density floor for 2D geometries and resolution;
- consistency between the cell-centered velocity used by momentum prediction and the face velocity used by the divergence constraint;
- behavior when several disconnected fluid fragments are present;
- conservation and normalization of fluid/VOID volume fractions after geometric transport; and
- the treatment of pressure and velocity at free-surface boundaries.

The current evidence supports changing the cell-centered pressure correction, not changing the active-cell classification or weakening the linear-solver convergence criterion.